

Twin Transformation: Digitalization and Verified Emissions

First findings — TWIN Analytics — github.com/FIMPlank/TWIN_Analytics — September 2026

Does a country's digitalization intensity predict a reduction in its independently verified (EU ETS) industrial emissions? Across two panels, five model specifications, three small-cluster-robust inference methods, and two rounds of adversarial review, no significant relationship was detected — at either the country or the industrial-sector level. The one earlier result that briefly looked significant ($p=0.047$, country-level, 3-year EIBIS panel) did not survive correction for the panel's true cluster structure, and does not reappear in an 11-year panel built on a different, independent digitalization measure.

The question

The EU's "twin transformation" framing assumes digital and green transitions reinforce each other. This project tests a narrower, checkable version of that claim: does digitalization intensity predict *verified* (independently audited) emissions reduction — or only disclosed climate ambition? This report covers the verified-emissions side only, at country and sector level (firm-level analysis needs a separate EIBIS microdata proposal to the EIB — not yet submitted; see Limitations).

Data and method

- **Verified emissions** — EU ETS / EUTL, country $\times$ activity $\times$ year, 2005–2025 (independently audited, not self-reported).
- **Digitalization** — two independent measures: Eurostat's Digital Intensity Index (country $\times$ NACE sector $\times$ year, 2015–2025, primary) and the EIB Investment Survey (country-level, 2023–2025, cross-check; $r=0.709$ with DII on the overlap — consistent, not independent corroboration).
- **Controls** — GDP growth, industrial electricity prices, energy-import dependency (energy-shock exposure), EU-accession cohort (catch-up/convergence proxy) — all from Eurostat's free dissemination API.
- **Two panels** — (a) country $\times$ year, 11 years, 29 countries; (b) country $\times$ ETS-sector $\times$ year, via a hand-built, confidence-rated NACE $\leftrightarrow$ ETS-activity crosswalk, reported on two cuts (full, and excluding the one low-confidence sector mapping) so no result rests on a single shaky merge.
- **Inference** — every coefficient reported under three methods (naive cluster-robust, $t(G-1)$, wild cluster bootstrap), plus a Granger-style placebo test (does future digitalization "predict" past emissions?) and a convergence-interaction test.

Quality process

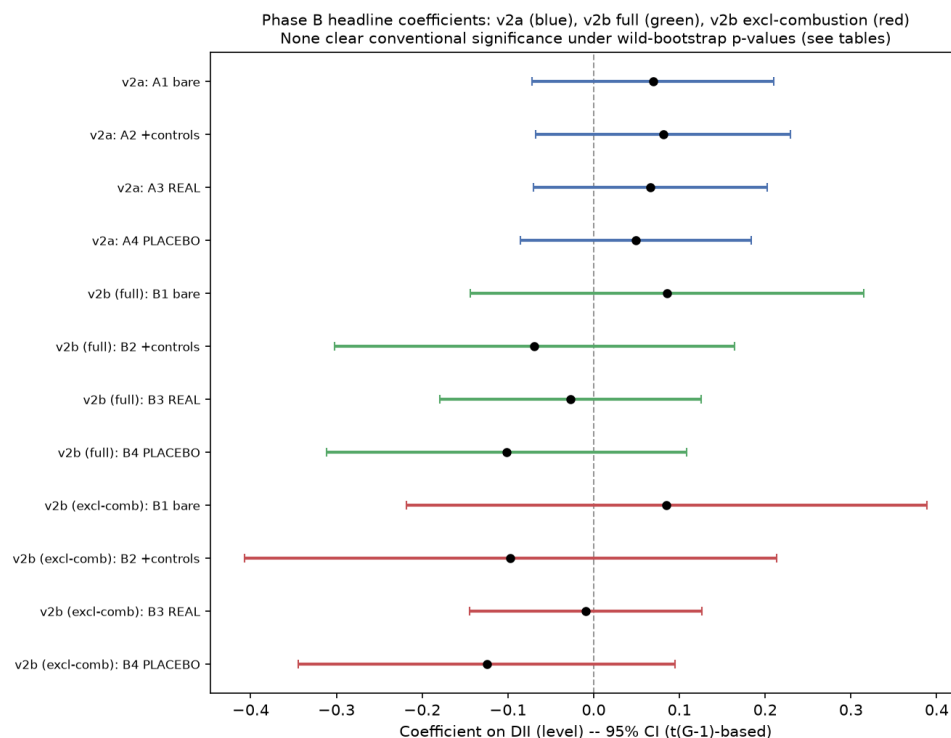
Every result below was produced by a Researcher agent and independently re-derived, line by line, by a separate Reviewer agent before being reported — across five review rounds. The Reviewer caught and the Researcher fixed: a backwards sign interpretation, uncorrected small-cluster standard errors that overstated one early result's significance by 5–30 $\times$, a mismatched-sample placebo comparison, and overclaiming in three other places. Nothing below is a first-draft number.

Key coefficients at a glance

Model	Panel	n	Clusters	Coef.	p (naive)	p (t, G–1)	p (wild bootstrap)
Bare (level + year FE)	Country, 2015–25	290	29	+0.069	0.314	0.323	0.358
+ controls (GDP, accession, energy)	Country, 2015–25	284	28	+0.081	0.263	0.272	0.291
Bare, full 7-sector cut	Sector, 2015–25	782	29	+0.085	0.447	0.453	0.489
Bare, excl.-combustion cut	Sector, 2015–25	697	28	+0.085	0.566	0.570	0.623
DII vs. EIBIS (same sample)	Country, 2023–25	79	27	+0.175 / +0.182	0.288 / 0.224	0.298 / 0.235	0.417 / 0.278

Coefficient = change in log verified emissions per 1-unit (0 $\rightarrow$ 100%) change in digitalization intensity, DII level (or EIBIS "multiple technologies" share, last row only). No p-value in this table, or anywhere in the full analysis, clears the conventional 0.05 threshold. Full model list (18+ specifications): [analysis/output/phase_b_v2a_results.csv](#), [phase_b_v2b_results.csv](#).

Headline result



DII coefficient (level) on year-over-year change in verified emissions, with 95% confidence intervals, across the four core specifications for the country panel (blue) and both sector-panel cuts (green = full, red = excluding the low-confidence combustion mapping). Every interval crosses zero; the minimum p-value across all 18+ models tested is 0.199 (naïve) / 0.199–0.89 depending on method — never close to conventional significance.

What the follow-up checks found

- **No reverse-causality artifact.** On a sample matched exactly between the lagged and led specifications, “future” digitalization does not predict past emissions any better than “past” digitalization predicts future emissions (country panel: +0.066 vs. +0.049, both non-significant).
- **The sector panel's sign instability traces to one variable, not noise.** Whether a country joined the EU in 2004 or later is specifically what flips the sector-level coefficient's sign (consistently across both crosswalk cuts) — the country panel's sign is untouched by it. This does not prove a catch-up-economics story, but it means that explanation cannot be ruled out, only that the data lacks the power (~28 country clusters) to confirm or reject it cleanly.
- **Two independent digitalization measures agree — on finding nothing.** DII (+0.175) and EIBIS (+0.182) give near-identical, both non-significant coefficients on the same 2023–2025 sample. Useful as a consistency check, not as two independent lines of evidence, since the measures themselves correlate at $r=0.709$.

Limitations

- Correlational and country/sector-level only — not causal, and not informative about whether the *same firms* that digitalize are the ones cutting emissions.
- Country-level digitalization coverage is genuinely short (DII: 11 years; EIBIS: 3 usable years) relative to the number of countries — statistical power to detect a small effect is limited throughout.
- The sector crosswalk is imperfect by construction: EU ETS's single largest category (combustion, ~57% of covered emissions) maps only loosely to a DII sector, which is why results are always shown both with and without it.
- No firm-level data yet — the EIBIS microdata proposal to the EIB (which pairs digitalization and climate variables on the same respondents) has not been submitted.

Bottom line

With real statistical power behind it — two independent digitalization measures, 11 years of data, sector-level variation, and adversarial review at every step — digitalization intensity shows no detectable relationship with verified industrial emissions at the country or sector level in the EU. That is itself informative: it rules out a *large* effect, and redirects the next round of work toward firm-level data, where an aggregation-diluted effect could still be hiding.